INCH-POUND

MIL-S-44220A(GL)
16 June 1989
SUPERSEDING
MIL-S-44220(GL)
31 March 1986

MILITARY SPECIFICATION

SLEEPING BAG, COLD WEATHER AIRCRAFT SURVIVAL KIT (VACUUM PACKED)

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a vacuum packed sleeping bag of one type and size for use in survival kits in aircraft operating in cold climates.

* 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

V-F-106	-	Fasteners, Slide, Interlocking
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8465

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:42Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-C-5040	-	Cord, Fibrous,Nylon
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-F-43097	-	Feathers and Down, Waterfowl; TAN-O-QUIL-QM Treated
MIL-T-43548	-	Thread, Polyester Core: Cotton-, Rayon-, or Polyester-Covered

STANDARDS

FEDERAL

FED-STD-595	-	Colors
FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members For Containers and Pallets
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:42Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Guide sample. Samples of the sleeping bag only (not vacuum packed), when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from this document may appear in the sample, in which case this document shall govern.

* 3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document.

3.3.1 Shell fabric. The inner and outer shell fabric shall be 1.9 ounces ± 0.2 ounce per yard ripstop, 70 denier nylon fabric woven with 109 yarns per inch minimum in the warp direction and 99 yarns per inch minimum in the filling direction. There shall be a minimum of five repeat yarns per inch of the reinforcing ribs in both the warp and filling directions. The material shall be water-repellent treated.

3.3.2 Baffling material. The baffling material shall be a 1 ± 0.1 ounce per square yard, 20 denier, single-warp-knit stitch tricot knit fabric with 190 yarns per inch minimum in the warp direction and 32 yarns per inch width in the filling direction.

* 3.3.3 Slide fastener. The slide fastener shall be size MHS, type III, style 1, nylon or polyester continuous element fastener and shall meet all the requirements of V-F-106. The fastener shall have a reversible double-pull slider with wire stirrup pulls and thongs attached, and the slider shall be free running for emergency exit. The slide fastener tape shall be 1 ± 1/16 inch width and water-repellent treated. The length of the slide fastener shall be a minimum of 65 inches, and all slider stops, tape, and chain components shall be black.

3.3.4 Cord, nylon. The nylon drawcord shall conform to type II of MIL-C-5040. The color shall be black.

3.3.5 Clamp, cord, plastic. The cord clamp shall consist of a lock body and lock wheel. The lock body shall conform to Fastex No. 238-390703-00-5612 or equal, and the lock wheel shall conform to Fastex No. 238-082051-00-0101 or equal. The color shall be black.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:42Z
Check the source to verify that this is the current version before use.

3.3.6 Thread. The thread for seaming and stitching shall be No. 50, 2 ply, conforming to MIL-T-43548.

3.3.7 Down. The down shall conform to type II, class 1 of MIL-F-43097, except that the TAN-O-QUIL-QM treatment and all related requirements and related inspection provisions need not apply.

3.3.8 Instruction sheets. The user instructions shall be legibly printed on two 8-1/2 by 11 inch white, bond paper sheets. The contents of each instruction sheet for opening and use shall be in accordance with figures 1 and 2 of this document.

* 3.3.8.1 Identification marking. Identification marking shall be placed in the upper left hand corner of the label shown on figure 1 of this document. This identification marking shall conform to class 5 of DDD-L-20.

3.4 Design. The sleeping bag shall be a mummy bag with a slide fastener closure and integral hood with drawstring closure with quick release for hood closure and be down filled.

3.5 Color. The color of the sleeping bag shall conform to International Orange No. 12246 of FED-STD-595.

3.6 Construction. The sleeping bag shall be differential-cut construction with all baffling seams, both internal and external, turned and finished toward the inside.

3.6.1 Insulation distribution. The sleeping bag shall be down insulated with the down evenly distributed throughout the bag with each channel (as formed by the baffling) containing equal and proportionate amounts of the total amount of down (see 6.6).

3.6.2 Stitches, seams, and stitching. Stitches, seams, and stitching shall conform to FED-STD-751. Stitch type shall be 301; seam type shall be SSa-1; and stitches per inch shall be 8-10.

3.6.2.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched 1/2 inch minimum, except when caught in other stitching or turned under in a hem. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread, or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

3.6.2.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:42Z
Check the source to verify that this is the current version before use.

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, and continue a minimum of 1/2 inch beyond the defective area on to the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs the ends of the stitching are not required to be backstitched.

3.6.2.2 Trimming. All thread ends shall be trimmed to a maximum length of 1 inch.

* 3.7 Use of automated equipment. Automated equipment may be used to perform any of the operations specified, providing that the seam and stitch type areas specified and the finished component conforms to the required configurations.

* 3.8 Finished measurements. The measurements of the finished sleeping bag shall be a minimum of 91 inches long by 32 inches wide at the widest point and 22 inches wide at the narrowest point. The dimensions shall be taken on the outside of the bag. The length shall be taken from the foot to the top of the hood in the open position. The width shall be taken at the widest point with the slide fastener closed and the bag laid out flat.

3.9 Vacuum packing. The sleeping bag shall be vacuum packed in two layers of a heat-sealable transparent film. Each layer of the film shall be fitted with a pull tab for easy opening. Both pull tabs shall be on the same side of the package. The vacuum packed sleeping bag shall be capable of passing the vibration, altitude, low temperature and ease of opening tests specified in 4.5.1, 4.5.2, and 4.5.3. The maximum dimensions of the vacuum packed sleeping bag shall be 13 by 13 by 2-1/2 inches. The instruction sheet for opening the package (figure 1) shall be placed between the two layers of film on the same side as the pull tabs so as to be readable before opening. The instructions for use (figure 2) shall be placed between the two layers of film on the side opposite the pull tabs.

3.10 Insulating "clo" value. After removal from the vacuum packing, the sleeping bag shall have a test sample average "clo" value of 7.1 with no individual "clo" value of less than 6.8 when tested as specified in 4.5.4.

3.11 Workmanship. The finished sleeping bag shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels. Vacuum packing shall not show any sign of loss of vacuum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:42Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested as specified in 4.4.5. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:42Z
Check the source to verify that this is the current version before use.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the component and material conforms to the requirements specified in 3.3.1 and 3.3.2. Each certificate of compliance shall be supported by attached test reports, ordering or receiving records, or other specifications supporting the statement of compliance.

4.4.2 In-process inspection. Inspection shall be made of the weight of down used in each sleeping bag and recorded. Records of inspection of the weight of sleeping bags in each lot shall be furnished to the Government as specified (see 6.6).

* 4.4.3 Sleeping bag visual and dimensional examination. The sleeping bag shall be examined for the defects listed in table I. The lot size shall be expressed in units of completed sleeping bags, prior to vacuum packing. The sample unit shall be one sleeping bag, prior to vacuum packing. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

TABLE I. End item visual and dimensional defects

Examine	Defect
Design	Sleeping bag not mummy style with integral hood
Slide fastener	Less than 65 inches long Does not have double reversible metal sliders Not free running Tape width not as specified
Color	Not as specified
Dimensions	Length or width not as specified
Construction and workmanship	Drawstring not as specified or missing Drawstring locking device not as specified or missing Any hole or tear in fabric Any piecing, splicing, or repair in materials Any thread breaks not overstitched

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:42Z
Check the source to verify that this is the current version before use.

TABLE I. End item visual and dimensional defects (cont'd)

Examine	Defect
Construction and workmanship (cont'd)	Any stitching ends not secured by at least 1/2 inch overlapping or backstitching Any thread ends not trimmed to 1 inch or less Any baffling seam not finished to interior Any open seam which will allow down to escape Any component missing or not as specified

4.4.4 Vacuum pack visual and dimensional examination. The vacuum packed sleeping bag shall be examined for the defects listed in table II. The lot size shall be expressed in units of vacuum packed sleeping bags. The sample unit shall be one vacuum packed sleeping bag. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

TABLE II. Vacuum pack visual and dimensional defects

Examine	Defect
Dimensions	Length, width, or depth exceeds maximum specified
Instructions	Either instruction sheet missing, incorrect, incomplete, illegible, or cannot be read without opening the vacuum pack
Construction and workmanship	Easy opening pull tab missing from either layer of transparent film Not packed in two layers of material Any visual indications of loss of vacuum, such as looseness of the transparent film

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:42Z
Check the source to verify that this is the current version before use.

4.4.5 End item vacuum pack testing. The vacuum packed sleeping bag shall be tested as specified in 4.5.1, 4.5.2, 4.5.3, and 4.5.4. The lot size shall be expressed in units of vacuum packed sleeping bags. The sample unit shall be one vacuum packed sleeping bag. The sample size shall be in accordance with the list below. Each sample unit shall be subjected to all tests and failure of any sample unit to pass all tests shall be cause for rejection of the lot.

Lot size	Sample size (sample units)
1 to 50	2
51 to 500	3
501 to 35000	5

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:42Z
Check the source to verify that this is the current version before use.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

* 4.5.1 Vibration test. Place a vacuum packed sleeping bag in the rigid survival seat kit of an OV-1 aircraft along with the other survival equipment included in the kit (see 6.5). Attach the survival seat kit solidly to a vibrating machine. Vibrate in accordance with method 514.3, procedure I of MIL-STD-810. The time schedule for each axis X, Y, and Z without vibration insulators shall be as follows:

a. Dwell time at top three resonance frequencies - 30 minutes.
b. Sinusoidal cycling time - 3 hours less dwell time and resonance search time.
c. Sweep time - Logarithmically over frequency range.
d. Frequency range - 5 to 500 to 5 HZ.
e. Vibration level:

Frequency	Acceleration	Displacement
Range (Hz)	(g)	(inches-double amplitude)
5-14	-	0.10
14-23	1	-
23-52	-	0.036
52-500	5	-

After completion of the vibration test the vacuum packed bag shall be removed from the kit and examined for any loss of vacuum indicated by looseness of the packing or any abraded holes or tears. Any damage to the sleeping bag pack shall constitute failure of the test.

4.5.2 Altitude and low temperature test. Measure the height of the vacuum packed sleeping bag. Place the vacuum packed sleeping bag in a temperature, altitude chamber along with a scale positioned in such a manner that any bag increase in dimension can be observed. Reduce the air pressure to simulate 25,000 feet and reduce the temperature to -20°F. Hold at these conditions for 2 hours. Periodically observe the operation through the chamber window to see

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:42Z
Check the source to verify that this is the current version before use.

if the sleeping bag has expanded in height due to reduced air pressure. At the end of this time increase the pressure to sea level, keep the temperature at -20°F, and hold for 1 hour. Next, remove the bag from the chamber, measure the height of the bag, and inspect the vacuum packing for any sign of temperature or altitude related damage. Any increase in the height of the bag (while in the chamber or after removal) or any damage to the packing material shall constitute failure of the test.

4.5.3 Vacuum packed sleeping bag opening test. After the vibration, altitude, and low temperature testing, the vacuum packed sleeping bag shall be cold soaked for 4 hours at -40°F. At the end of this time, the vacuum packed sleeping bag shall be opened by pulling firmly with one hand in accordance with the instructions provided. The pull tabs shall break the packing layer to which they are affixed when pulled. If the tabs do not break the layer as required, it shall be considered a failure.

4.5.4 Insulation "clo" value test. After removal from the vacuum pack, the sleeping bag shall be tested for "clo" value using the single circuit Copper Man G.E. Cat PX15A4 (see 6.7). Inability of any individual sleeping bag to meet the minimum insulating characteristic, or inability to meet the test sample average insulating characteristics shall constitute failure of the test.

* 4.5.4.1 Experimental conditions.

a. Mannequin surface temperature of 90 to 95°F, for the body and 85 to 90°F for the hands and feet.

b. Ambient temperatures (measured at the same height as the mannequin) one foot from the bag to produce a positive delta temperature between the ambient and the mannequin of at least 20°F.

c. Air movement controlled at 50 to 60 ft/min. At least one measurement shall be made at 1 mph horizontal flow, parallel with the long axis of the sleeping bag. The bag shall be placed with the head in the direction of the flow (e.g., windward). Air flow should be uniform within the space of the chamber.

d. Relative humidity (r.h.) shall be controlled with deviations of dew point temperature, not to exceed 1°C or ± 3 percent r.h., whichever is less.

4.5.4.2 Sizes and dressing details.

a. Bag size as provided.

b. Mannequin dressed in 50 percent cotton, 50 percent wool pajama-type underwear, standard issue, regular medium size, and standard wool socks with cushion soles.

c. Fluff bag by shaking violently for 2 minutes prior to placing mannequin in bag.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:42Z
Check the source to verify that this is the current version before use.

d. Position mannequin so that sides and ends of bag are not compressed. Secure all closures. Cover mannequin completely, including face. If bag cannot be drawn over face, cover with 1 inch of wool pile insulation.

* 4.5.4.3 Operation procedures.

a. Position bag and mannequin in center of elevated non-metallic support plate with an open weave lattice-type surface. The open weave surface will ensure uniform air flow on all exposed surfaces of the bag.

b. Equilibrate system until all parameters are at steady state and the power has not been more than ± 2 percent for at least one half hour.

c. Make three separate measurements of insulation value. Record mannequin and air temperatures, mannequin power input, relative humidity and air velocity.

* 4.5.4.4 Calculations.

a. Calculate insulating value of the bag, including boundary air insulation.

$$clo = \frac{3.09\ (T_s - T_a)\ (SA)}{power}$$

Where:

T_s = surface temperature of mannequin (degrees F)
T_a = ambient air temperature, dry bulb (degrees F)
SA = surface area (m) of the respective mannequin
power = watts

* 4.5.4.5 General. The contractor shall report the type of measuring apparatus used (thermocouple, thermister, or other type) and briefly describe the system used (locations on mannequin where measurements are taken).

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

* 5.1.1 Level A packing. Ten sleeping bags shall be packed flat, two in length, one in width, and five in depth in a snug-fitting fiberboard shipping container conforming to style FOL-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Inside dimensions of each shipping container shall be approximately 26-1/2 inches in length, 13-3/4 inches in width, and 11-1/2 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:42Z
Check the source to verify that this is the current version before use.

accordance with method III and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

* 5.1.2 Level B packing. Ten sleeping bags shall be packed flat, two in length, one in width, and five in depth in a fiberboard shipping container conforming to style FOL-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Inside dimensions of each shipping container shall be approximately 26-1/2 inches in length, 13-3/4 inches in width, and 11-1/2 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II, as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.1.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.1.3 Commercial packing. Sleeping bags shall be packed in accordance with ASTM D 3951.

* 5.2 Palletization. When specified (see 6.2), sleeping bags, packed as specified in 5.1.2 and 5.1.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 6 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The vacuum packed sleeping bags are intended for use in cold weather aircraft survival kits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:42Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Level of packing (see 5.1).
e. Type and class of unit load (see 5.1.1).
f. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.1.2.1).
g. When palletization is required (see 5.2).
h. Directions for control and use of the survival seat kit (see 4.5.1 and 6.5).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Samples. For access to samples, address the contracting activity issuing the invitation for bids.

6.5 Survival seat kit. The OV-1 survival seat kit required for the vibration test in 4.5.1 should be furnished by the procuring activity.

6.6 Weight of down. A minimum weight of 50 ounces of down, dispersed evenly throughout the sleeping bag has been found to provide a sleeping bag with environmental protection meeting the requirements of this specification. This weight is provided for information only and does not relieve the contractor from complying with all requirements of the contract and this specification.

6.7 Approved insulation "clo" value testing facilities. The following are the only approved testing facilities for the insulation "clo" value test (see 4.5.4). Use of any other facility without prior approval by the Government contracting officer shall constitute failure of the test and rejection of any material represented by non-approved test results.

Kansas State University
Manhattan, Kansas 66502

U.S. Army Research Institute of Environmental Medicine
Kansas Street
Natick, MA 01760-5014

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:42Z
Check the source to verify that this is the current version before use.

* 6.8 Subject term (key word) listing.

Aircraft, life support
Equipment, aircraft life-sustaining
Equipment, cold weather

6.9 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL

Review activities:

Army - MD
DLA - CT

Preparing activity:

Army - GL

(Project 8465-A008)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:42Z
Check the source to verify that this is the current version before use.

SLEEPING BAG, ARCTIC SURVIVAL

TO OPEN:

HOLD PACKAGE FIRMLY
PULL TAPE AS SHOWN
REPEAT ON INNER BAG

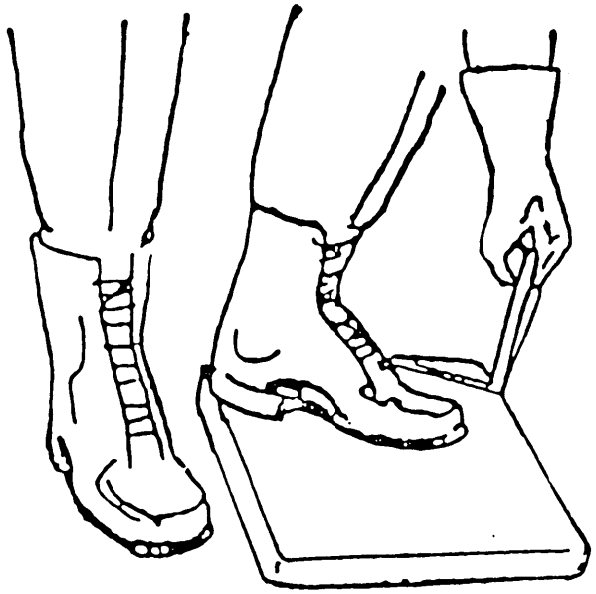


FIGURE 1. Sleeping bag, Arctic survival.

Source: https://assist.dla.mil -- Downloaded: 2026-02-22T00:42Z
Check the source to verify that this is the current version before use.

USER INSTRUCTIONS

1. KEEP THE BAG DRY:

a. SELECT THE DRIEST GROUND AND IF POSSIBLE, KEEP THE BAG OUT OF THE RAIN.

b. BREATHE THROUGH FACE OPENING TO PREVENT MOISTURE FROM WETTING THE BAG. IF FACE IS COLD, REDUCE FACE OPENING BY TIGHTENING UP DRAWSTRINGS WITH THE PLASTIC CORDLOCK.

c. DO NOT WEAR DAMP CLOTHING AND AVOID SWEATING IN BAG. IF TOO WARM, OPEN ZIPPER FOR VENTILATION.

d. OPEN THE BAG COMPLETELY AND AIR THOROUGHLY EACH DAY.

e. IF POSSIBLE, USE A GROUND CLOTH UNDER THE SLEEPING BAG. A LIFE RAFT OR PARACHUTE CAN BE USED FOR THIS, IF AVAILABLE.

2. TO KEEP WARM:

a. FLUFF BAG THOROUGHLY BEFORE USING.

b. IF POSSIBLE, GIVE THE BAG ENOUGH TIME TO LOFT OR EXPAND AFTER IT HAS BEEN FLUFFED.

3. PRECAUTIONS:

a. BE CAREFUL OF SPARKS FROM CAMPFIRES LANDING ON THE SLEEPING BAG.

b. DO NOT SMOKE IN THE SLEEPING BAG.

c. AVOID TEARING OF SLEEPING BAG SHELL BY DRAGGING ACROSS SHARP OBJECTS SUCH AS BRANCHES, STONES, OR SURVIVAL EQUIPMENT. DOWN LEAKAGE WILL OCCUR WITH SUBSEQUENT LOSS IN INSULATING PROPERTIES.

4. ZIPPER OPERATION:

a. WHEN CLOSING ZIPPER, CARE MUST BE TAKEN TO PREVENT THE NYLON SHELL MATERIAL FROM BEING CAUGHT IN ZIPPER TEETH.

5. SECONDARY METHOD OF ENTRY:

a. THE SLEEPING BAG CAN BE ENTERED FROM THE TOP OPENING IF THE ZIPPER BECOMES JAMMED SHUT.

FIGURE 2. Sleeping bag, user instructions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:42Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS. In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

US Army Natick Research, Development, and
Engineering Center
Natick, MA 01760-5014

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300
STRNC-ES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Natick Research, Development,
and Engineering Center
ATTN: STRNC-ES
Box 14A
Natick, MA 01760-5014

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:42Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-44220A(GL)	Sleeping Bag, Cold Weather Aircraft Survival Kit (Vacuum Packed)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:42Z
Check the source to verify that this is the current version before use.